

Formulaire de trigonométrie

Identités fondamentales

$$\cos^2(x) + \sin^2(x) = 1$$

$$1 + \cot^2(x) = \frac{1}{\sin^2(x)}$$

$$1 + \tan^2(x) = \frac{1}{\cos^2(x)}$$

Formules d'addition

$$\cos(a + b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$\sin(a + b) = \sin(a) \cos(b) + \cos(a) \sin(b)$$

$$\tan(a + b) = \frac{\tan(a) + \tan(b)}{1 - \tan(a) \tan(b)}$$

$$\cos(a - b) = \cos(a) \cos(b) + \sin(a) \sin(b)$$

$$\sin(a - b) = \sin(a) \cos(b) - \cos(a) \sin(b)$$

$$\tan(a - b) = \frac{\tan(a) - \tan(b)}{1 + \tan(a) \tan(b)}$$

Formules de duplication

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\tan(2x) = \frac{2 \tan(x)}{1 - \tan^2(x)}$$

$$\cos(2x) = \cos^2(x) - \sin^2(x) = 2 \cos^2(x) - 1$$

Formules d'angle moitié

$$\sin^2\left(\frac{x}{2}\right) = \frac{1 - \cos(x)}{2}$$

$$\tan\left(\frac{x}{2}\right) = \frac{\sin(x)}{1 + \cos(x)} = \frac{1 - \cos(x)}{\sin(x)}$$

$$\cos^2\left(\frac{x}{2}\right) = \frac{1 + \cos(x)}{2}$$

Produit en somme

$$\cos(a) \cos(b) = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$$

$$\sin(a) \cos(b) = \frac{1}{2} [\sin(a + b) + \sin(a - b)]$$

$$\sin(a) \sin(b) = \frac{1}{2} [\cos(a - b) - \cos(a + b)]$$

Somme en produit

$$\cos(a) + \cos(b) = 2 \cos\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

$$\sin(a) + \sin(b) = 2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

$$\cos(a) - \cos(b) = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$$

$$\sin(a) - \sin(b) = 2 \cos\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$$

Formules liées aux angles remarquables

$$\cos(\pi - x) = -\cos(x)$$

$$\tan(\pi - x) = -\tan(x)$$

$$\cos(\pi + x) = -\cos(x)$$

$$\tan(\pi + x) = \tan(x)$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cotan(x)$$

$$\cos\left(\frac{\pi}{2} + x\right) = -\sin(x)$$

$$\tan\left(\frac{\pi}{2} + x\right) = -\cotan(x)$$

$$\sin(\pi - x) = \sin(x)$$

$$\sin(\pi + x) = -\sin(x)$$

$$\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$$

$$\sin\left(\frac{\pi}{2} + x\right) = \cos(x)$$